



Homogeneity of peraluminous $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Al}_2\text{O}_3\text{--Na}_2\text{O--CaO--Nd}_2\text{O}_3$ glasses: Effect of neodymium content

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ARTICLE INFO

Article history:

Received 22 May 2014

Received in revised form 17 July 2014

Available online 14 September 2014

Keywords:

Aluminoborosilicate glass;

Peraluminous;

Neodymium oxide;

Glass crystallization;

Phase separation

ABSTRACT

Considering the interest of developing new glass matrices able to immobilize higher concentration of high level nuclear wastes than currently used nuclear borosilicate compositions, glasses containing high rare earth contents are of particular interest. This study focuses on a peraluminous aluminoborosilicate system $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Al}_2\text{O}_3\text{--Na}_2\text{O--CaO--Nd}_2\text{O}_3$ defined by a peralkaline/peraluminous ratio $R_p = ([\text{Na}_2\text{O}] + [\text{CaO}]) / ([\text{Na}_2\text{O}] + [\text{CaO}] + [\text{Al}_2\text{O}_3]) < 0.5$. Samples with various contents of Nd_2O_3 from 0 to 10 mol% were studied using DSC, XRD, SEM, TEM, STEM and EMPA methods. The glasses present a high thermal stability even after a slow cooling treatment from the melt. Only a slight mullite crystallization is detected for low Nd_2O_3 content (≤ 2.3 mol%) and crystallization of a neodymium borosilicate crystalline phase combined to a phase separation occurred at high Nd_2O_3 content (≥ 8 mol%). The solubility of neodymium in the presence of aluminum is demonstrated, with higher neodymium incorporation amounts than in peralkaline glasses.

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1. Introduction

Reprocessing of the spent nuclear fuel generates high level nuclear wastes which contain fission products and minor actinides that are currently immobilized in a borosilicate glass in France, called R7T7 glass. In order to improve the reactor efficiency, an increase of the combustion rate could occur in the near future, meaning that higher amounts of fission products and especially of minor actinides would be generated. To limit the number of canisters produced per ton of used fuel, an increase of the waste incorporation rate in nuclear glass might be considered [1]. The actual R7T7 glass matrix, intended for a maximum waste loading of 18.5 wt.%, provides optimal thermal stability and long-term chemical durability [2]. However, increasing the waste content could lead to higher thermal outputs and subsequent crystallization. In this context, current research focuses on new glass formulations able to incorporate higher quantities of fission products and minor actinides, with enhanced properties in terms of glass homogeneity and long-term behavior. Homogeneous glasses in either the melt or after cooling of the canister are required.

Glasses with high rare earth content appear of particular interest, since they constitute a significant part of the waste composition and are usually used as actinide surrogates in non-radioactive environment [3,4].

Moreover, aluminum-rich compositions have shown that the presence and the content of Al have a strong influence on the rare earth solubility in ternary systems such as $\text{SiO}_2\text{--Al}_2\text{O}_3\text{--RE}_2\text{O}_3$ or $\text{B}_2\text{O}_3\text{--Al}_2\text{O}_3\text{--La}_2\text{O}_3$ [5–9] or in more complex systems such as $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Na}_2\text{O--Al}_2\text{O}_3$ [10–14] or $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Na}_2\text{O--Al}_2\text{O}_3\text{--CaO--ZrO}_2$ [15–21]. In these studies, a peralkaline/peraluminous R_p ratio has been defined as $R_p = ([\text{M}'\text{O}] + [\text{M}_2\text{O}]) / ([\text{M}'\text{O}] + [\text{M}_2\text{O}] + [\text{Al}_2\text{O}_3])$ with M and M' corresponding to alkaline and alkaline-earth elements respectively. This parameter allows distinguishing a peralkaline glass (R_p superior to 0.5) where the modifier ions are in excess compared to aluminum ions from a peraluminous glass (R_p inferior to 0.5), where the modifier ions are lacking. It appears that the lanthanide solubility as well as the crystallization tendency of glasses strongly depends of this R_p value.

According to Li et al., in $\text{SiO}_2\text{--B}_2\text{O}_3\text{--Na}_2\text{O--Al}_2\text{O}_3\text{--RE}_2\text{O}_3$ peralkaline glasses ($R_p > 0.5$) the solubility limit of rare earth oxides decreases dramatically from about 8 mol% to less than 1 mol% when R_p decreases [11]. Above the solubility limit, crystallization of a sodium apatite $\text{Na}_x\text{RE}_{9.33-x}(\text{SiO}_4)_6\text{O}_2$ phase occurred. The authors suggested that these glasses were constituted by borate and silicate regions, with a

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Table 1

Nominal target compositions (mol%) of the G0.26Nd_x and G0.40Nd_x glasses at $R_p = 0.38$, with $[B_2O_3]/[SiO_2]$ respectively fixed at 0.26 and 0.40. x corresponds to the molar fraction of Nd₂O₃.

Glass	SiO ₂	B ₂ O ₃	Al ₂ O ₃	Na ₂ O	CaO	Nd ₂ O ₃
G0.26Nd0	57.45	14.90	17.02	6.38	4.25	0.00
G0.26Nd2.3	56.13	14.56	16.64	6.24	4.16	2.27
G0.26Nd4	55.10	14.29	16.33	6.12	4.08	4.08
G0.26Nd6	54.00	14.00	16.00	6.00	4.00	6.00
G0.26Nd8	52.85	13.71	15.66	5.87	3.91	8.00
G0.26Nd10	51.70	13.41	15.32	5.74	3.83	10.00
G0.40Nd0	51.49	20.85	17.02	6.38	4.25	0.00
G0.40Nd2.3	50.31	20.37	16.64	6.23	4.16	2.29
G0.40Nd4	49.39	20.00	16.33	6.12	4.08	4.08
G0.40Nd6	48.40	19.60	16.00	6.00	4.00	6.00
G0.40Nd8	47.38	19.18	15.66	5.87	3.91	8.00
G0.40Nd10	46.34	18.77	15.32	5.74	3.83	10.00

nanoscale phase separation. Similarly, other studies on the peralkaline SiO₂–B₂O₃–Na₂O–Al₂O₃–CaO–ZrO₂–RE₂O₃ system show that the more the R_p ratio decreased up to 0.5, the more the crystallization tendency of a calcium apatite Ca₂RE₈(SiO₄)₆O₂ phase increased after a slow cooling thermal treatment of 1 °C min^{−1} [4,15,17,20]. As proposed by Caurant et al. [22], in the presence of Na and Ca, the Ca-apatite would be thermodynamically more stable than Na-apatite [15,16,18]. Moreover in these studies, neither metaborate-like structure nor evidence of separated borate and silicate regions was put forward.

On the contrary, for *peraluminous* SiO₂–B₂O₃–Na₂O–Al₂O₃–RE₂O₃ glasses, the lower the R_p is, the higher the rare earth element content dissolved in the network is [10,12]. Nevertheless, a minimum amount of rare earth elements is required to get homogeneous glasses, depending on the glass composition [12,23]. For compositions with high Al₂O₃ concentration ($R_p < 0.35$) and low RE₂O₃ amount, crystallization of mullite (Al₄SiO₈)_{1.2} was noticed [23]. In the homogeneous domain – as for peralkaline glasses – Li et al. suggested that lanthanides were first incorporated in borate regions with a double chain structure similar to a lanthanide metaborate REB₃O₆ crystal before being incorporated in the silicate regions, leading to the crystallization of a NaRE₉Si₆O₂₆ apatite phase above the RE solubility limit.

In this study, new peraluminous matrices of the SiO₂–B₂O₃–Na₂O–Al₂O₃–CaO–Nd₂O₃ system with increasing rare earth contents, at two $[B_2O_3]/[SiO_2]$ ratios, were studied in terms of homogeneity, with a particular focus on the crystallization or phase separation tendency of

the glass. The aim of this work was to determine the homogeneity domains of peraluminous glasses and the lanthanide influence on the crystallization and phase separation tendencies in these samples. The influence of the peraluminous glass composition on the lanthanide solubility and glass homogeneity was investigated at two different cooling rates: quenching in air and slow cooling (1 °C min^{−1}), the latter to simulate the cooling rate of the glass in the bulk of the canister.

The homogeneity of the glasses was controlled by XRD, SEM and TEM analyses. When crystallization occurred, electron microprobe analysis was used to identify the crystal composition. In the case of phase separation, STEM analysis allowed to qualitatively determine the composition of separated phases.

2. Experimental

2.1. Glass preparation

Two series of six-oxide glasses were prepared in the SiO₂–B₂O₃–Al₂O₃–Na₂O–CaO–Nd₂O₃ system, using a fixed R_p value (0.38) with various neodymium contents: 0, 2.3, 4, 6, 8, and 10 mol%. The first one was prepared with a $[B_2O_3]/[SiO_2]$ molar ratio of 0.26, and the second one with a $[B_2O_3]/[SiO_2]$ ratio of 0.40, respectively named G0.26Nd_x and G0.40Nd_x in the following ($x = [Nd_2O_3]$). Compositions are summarized in Table 1.

All glasses were synthesized starting from the reactants SiO₂ (Millisil, purity 0.997), H₃BO₃ (purity 0.995), Al₂O₃ (purity 1.000), Na₂CO₃ (purity 0.998), CaO (purity 0.995), Nd₂O₃ (purity 0.999) (Alfa Aesar), dry mixed (150 g batches) and melted in Pt–Au crucibles at 1350 °C for a step of 3 h. Glasses were then quenched by pouring over a steel plate, milled down (planetary mill, Retch) to less than 400 μm in agate containers and melted again at 1350 °C for a step of 2 h to ensure homogenization of the melt before being poured in the same way as previously to obtain glasses for so called quenched samples.

A controlled slow cooling at 1 °C min^{−1} from 1350 °C to room temperature (simulating the natural cooling of the bulk of waste glass canisters) was also performed on 50 g of all quenched samples. Slow cooled samples are noted SC in the following.

The electron microprobe chemical analyses conducted confirmed that the composition of the final materials was within the expected values, especially for the volatile species, boron and sodium.

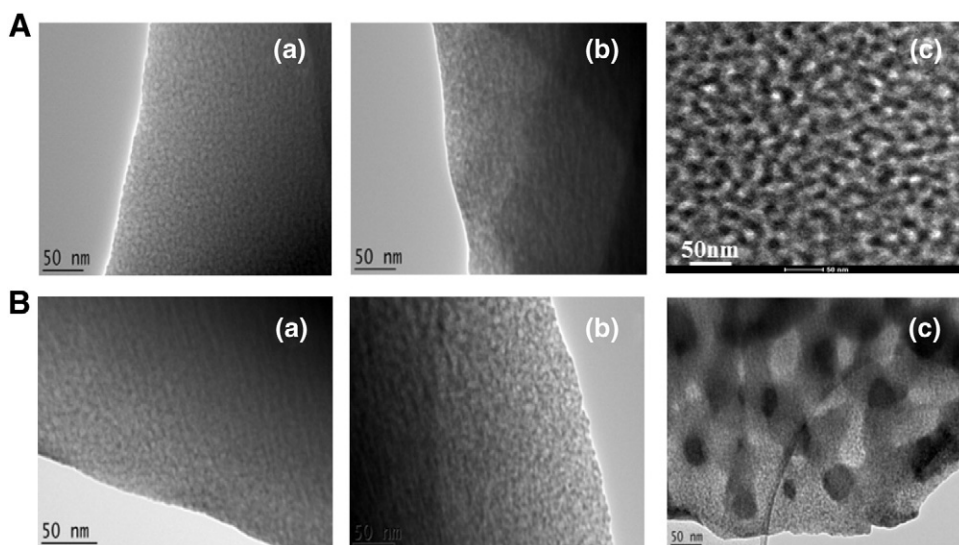


Fig. 1. TEM images of (A) quenched glasses and (B) slowly cooled glasses for the G0.26Nd_x series with the following Nd₂O₃ content: (a) 2.3, (b) 6 and (c) 10 mol%.

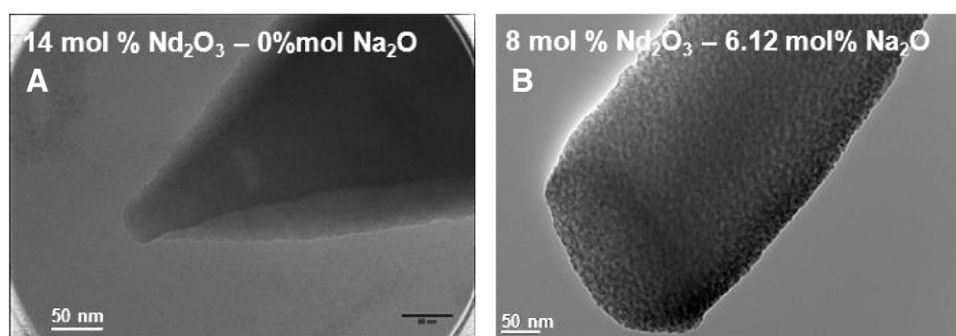


Fig. 2. TEM images of quenched glass (A) $49.39\text{SiO}_2\text{--}20.00\text{B}_2\text{O}_3\text{--}16.33\text{Al}_2\text{O}_3\text{--}14.28\text{La}_2\text{O}_3$. (B) $\text{G}0.26\text{Nd}8$ observed after a few minutes under the electronic beam.

2.2. Characterization methods

Differential Scanning Calorimetry (DSC) analyses were performed on a Multi HTC Setaram under argon atmosphere. Each sample ($m = 255 \text{ mg}$) was introduced in a Pt crucible and heated at a $10^\circ\text{C min}^{-1}$ rate and an empty crucible was used as reference.

Powder X-ray diffraction was carried out on a Philips X'pert Pro PANalytical instrument (Ni-filtered $\text{Cu K}\alpha$ source $\lambda = 1.5406 \text{ \AA}$) in a Bragg–Brentano geometry. X-ray diffraction patterns were recorded between 10 and 130° (2θ) (step $= 0.0174^\circ$). Modelization of the X-ray pattern was performed in two steps using the Fullprof_Suite program [24]. Lattice parameters were first determined by a whole pattern decomposition [25] then the structure was successfully modeled by the Rietveld technique [26].

Scanning Electron Microscopy (SEM) analyses were carried out on a Philips XL30 (15 kV, working distance: 11.3 mm) equipped with an energy dispersive X-ray (EDX) probe. Samples were first polished ($1 \mu\text{m}$) then observed with optical microscope Bruker before being carbon coated.

Electron microprobe analysis (EMPA) was carried out with a CAMECA SX50 electron probe microanalyzer equipped with four WDS spectrometers (PC3 (B), TAP (Na, Si, Al), PET (Ca) and LIF (Nd) crystals). An acceleration voltage of 15 kV, and a stabilized beam current of 10 nA were used. The probe diameter was set to $1 \mu\text{m}$ and to $10 \mu\text{m}$ for Na analysis.

Transmission Electron Microscopy (TEM) observations were performed on a Philips CM20 fitted out with an Oxford EDS analyzer. A drop of ethanol containing the small particles in suspension was deposited onto a carbon coated copper grid. Selected area electron diffraction (SAED) was also performed to check the presence of

crystalline phase and energy dispersive spectroscopy EDS was performed on samples which exhibited phase separation larger than 50 nm , taking into account the beam size. It should be noticed that due to the currently mentioned sodium migration induced by the beam irradiation, we limited the analysis duration to a few seconds in order to avoid sample change [27].

Scanning Transmission Electron Microscopy (STEM) analyses were performed on a FEI Tecnai G2 microscope equipped with a LaB_6 filament and a X-Max Oxford Silicon Drift Detector. The beam diameter was about 5 nm in this mode. Sample thickness obtained by Focused Ion Beam (FIB) cutting was about 100 nm . Profile measurements and semi-quantitative analyses were performed but electron radiation damages limited the precision of the measures.

In the case of phase separation, the boron distribution was investigated by elemental mapping obtained with Energy Filtered Transmission Electron Microscopy (EFTEM). This analysis was performed on a JEOL 2010 F TEM equipped with a Gatan Imaging Filter (GIF).

3. Results

3.1. Quenched glasses

After casting, all the quenched glasses appeared translucent (taking into account a slight violet coloration) and macroscopically homogeneous. No crystalline phase was detected on XRD diagrams for both of the $\text{G}0.26\text{Nd}x$ and $\text{G}0.40\text{Nd}x$ series. Back-scattered electron SEM observations confirmed the homogeneity of the samples at the micrometer scale. Only the $\text{G}0.26\text{Nd}0$ glass presented

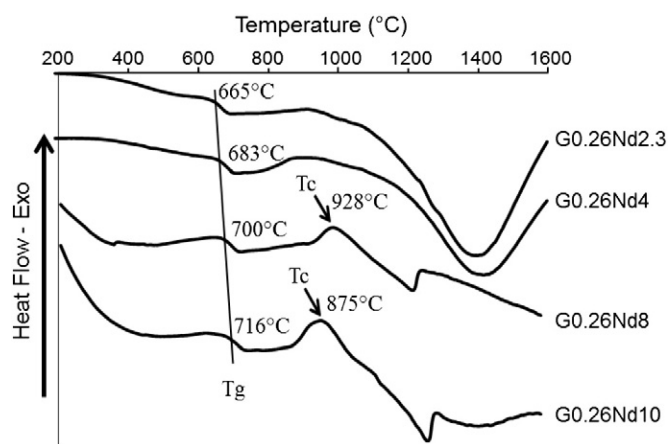


Fig. 3. Differential Scanning Calorimetry curves of the $\text{G}0.26\text{Nd}x$ glasses ($x = 2.3, 4, 8, 10$) (heating rate: $10^\circ\text{C min}^{-1}$, atmosphere: argon). The temperature values are estimated with an error of $\pm 1^\circ\text{C}$.

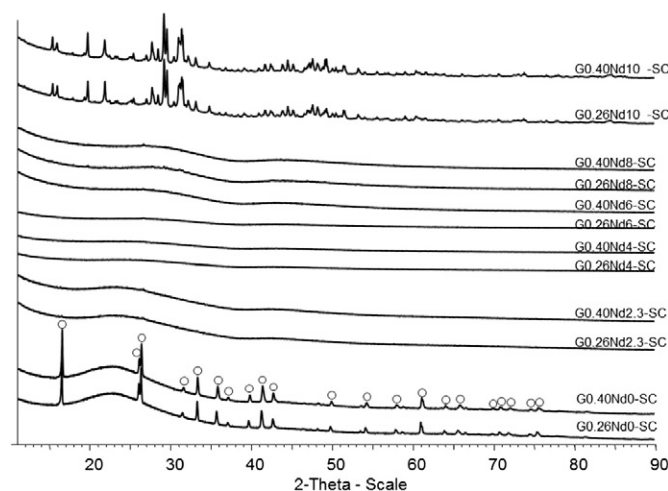


Fig. 4. XRD diagrams of the slowly cooled $\text{G}0.26\text{Nd}x\text{-SC}$ and $\text{G}0.40\text{Nd}x\text{-SC}$ glasses containing 0, 2.3, 4, 6, 8 and 10 mol% Nd_2O_3 . The $3\text{Al}_2\text{O}_3\text{--}2\text{SiO}_2$ mullite structure (ICCD-card: 15-0776) is indicated.

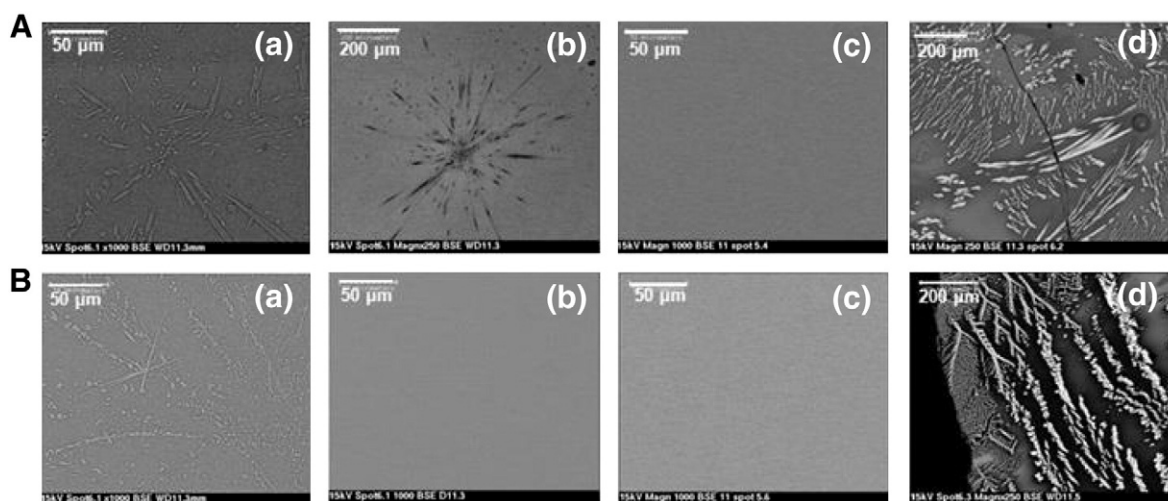


Fig. 5. Back-scattered SEM images of slowly cooled (A) G0.26Nd_x-SC and (B) G0.40Nd_x-SC glasses with the following Nd₂O₃ content: (a) 0, (b) 2.3, (c) 4 and (d) 10 mol%.

few dispersed crystals mainly composed of Al, Si, O according to EDS analysis.

According to TEM observations, no crystallization or phase separation was detected for the G0.26Nd_x and G0.40Nd_x glasses containing up to 8 mol% Nd₂O₃ (Fig. 1A). Note that the slight graininess is due to rapid sodium migration under the electronic beam. This effect is clearly visible from the beginning of the electron beam irradiation whereas no evolution was observed for a glass of composition 49.39SiO₂–20B₂O₃–16.33Al₂O₃–14.28La₂O₃, containing no sodium (Fig. 2). For G0.26Nd10 and G0.40Nd10 glasses, TEM and STEM analyses revealed a phase separation between two amorphous phases, dark and bright contrasted, forming droplets of about 20 nm diameter (Fig. 1A (c)). DSC analyses of the G0.26Nd_x series showed a strong impact of the increasing lanthanide content on the glass transition temperature T_g , which increased continuously from 665 °C for the G0.26Nd2.3 sample to 716 °C for the G0.26Nd10 sample (Fig. 3). As it was already observed for borate and alkaline silicate glasses [28], this evolution could be related to the increasing mean field strength of the modifier ions in the network. Moreover in the case of the G0.26Nd8 and G0.26Nd10 samples, an exothermic peak was respectively observed at 928 °C and 875 °C, corresponding to a crystallization process.

3.2. Slowly cooled glasses

After a slow cooling at 1 °C min^{−1}, the G0.26Nd0-SC and G0.40Nd0-SC samples appeared visually opaque, whereas the G0.26Nd2.3-SC sample seemed translucent with small crystals homogeneously dispersed in the matrix. Glasses containing from 4 to 6 mol% Nd₂O₃ for G0.26Nd_x-SC series and from 2.3 to 6 mol% Nd₂O₃ for G0.40Nd_x-SC series appeared macroscopically homogeneous. With 8 mol% Nd₂O₃, heterogeneous crystallization was observed at the glass/crucible interface for both series, whereas the bulk appeared homogeneous. Crystallization and opaque areas were clearly visible in the whole G0.26Nd10-SC and G0.40Nd10-SC glasses.

XRD diagrams of the slowly cooled glasses are shown in Fig. 4. Results are similar for both G0.26Nd_x and G0.40Nd_x series. The Bragg

peaks detected for the crystallized G0.26Nd0-SC and G0.40Nd0-SC samples could be assigned to a mullite structure close to the 3Al₂O₃–2SiO₂ mullite compound (ICCD-card 15-0776). Then, slow cooled glasses appeared amorphous from 2.3 to 8 mol% Nd₂O₃ according to XRD analysis. Finally, for both G0.26Nd10-SC and G0.40Nd10-SC samples, crystallization peaks are clearly visible.

Back-scattered electron SEM pictures are presented in Fig. 5. The G0.26Nd0-SC and G0.40Nd0-SC samples exhibit homogeneously dispersed crystals. Few of these crystals were also present as isolated heaps in G0.26Nd2.3-SC. The relative atomic proportions determined by EMPA (Table 2) are coherent with a boron-substituted mullite phase, as suggested by XRD analysis. G0.40Nd2.3-SC glass as well as glasses containing 4 to 6 mol% Nd₂O₃ of both series appeared homogeneous at the micrometer scale after slow cooling. Beyond a Nd₂O₃ content of 8 mol%, crystals with a bright contrast were systematically detected, at the glass/crucible interface for 8 mol% Nd₂O₃ of both series and in the whole samples at 10 mol%. Their shapes are various from needles of 200 μm length to more complex shapes of 20–50 μm size. A quantitative electron microprobe analysis of the crystals indicates the main presence of Nd, Si, B, and O, as well as a low content of Ca (inferior to 0.5 at.%) (Table 3). Some of these analyses revealed a slight presence of Al (value in the range of apparatus sensibility). Fig. 6 shows the evolution of Si, Al, B, Nd and Ca contents through a crystal surrounded by residual glass on the G0.26Nd10-SC sample (white line on the SEM image). This profile clearly shows that the crystals appeared largely enriched in Nd and depleted in Al compared to the surrounding matrix. A slight decrease of Ca, Si and B rates can also be noticed in the crystals.

Moreover, macroscopically opalescent areas were detected at 8 and 10 mol% for both G0.26Nd_x-SC and G0.40Nd_x-SC series. TEM and STEM analyses of the G0.26Nd8-SC and G0.40Nd8-SC samples revealed a phase separation between two interlaced amorphous phases, respectively dark and bright contrasted. The size of the domains increases with [B₂O₃]/[SiO₂] ratio, from 20 nm for G0.26Nd8-SC to 80 nm for

Table 2

EMPA composition of the crystals present in the G0.26Nd2.3-SC glass (mean value of 18 analyses). Oxygen content was determined in order to fix the sum of all elements at 100%.

Sample	Crystal composition (at.%)				
	Al	Si	B	Nd	O
G0.26Nd2.3-SC	27.2 ± 1.2	5.7 ± 0.7	5.8 ± 0.5	0.06 ± 0.03	61.2

Table 3

EMPA compositions of the crystals present in the G0.26Nd8-SC, G0.26Nd10-SC, and G0.40Nd10-SC samples (mean value of 15 analyses). B/Si ratios of the crystals are also indicated. Oxygen content was determined in order to fix the sum of all elements at 100%.

Sample	Crystal composition (at. %)					
	Nd	Si	B	Ca	O	B/Si
G0.26Nd8-SC	16.4 ± 0.5	11.5 ± 0.5	6.0 ± 0.6	0.3 ± 0.04	65.8	0.52
G0.26Nd10-SC	15.6 ± 0.4	11.4 ± 0.3	5.3 ± 0.4	0.3 ± 0.04	67.4	0.46
G0.40Nd10-SC	16.6 ± 0.5	11.3 ± 0.6	6.6 ± 0.6	0.3 ± 0.06	65.3	0.58
Theoretical Nd ₃ Si ₂ BO ₁₀	18.8	12.5	6.3	–	62.5	0.50

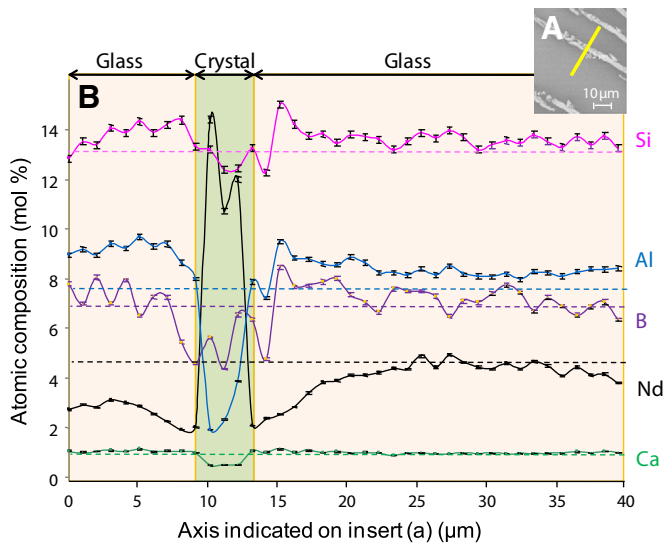


Fig. 6. Element profile measurement performed by electron microprobe analysis through a crystal and the surrounding residual glass for G0.26Nd10-SC sample. (A) SEM image showing the line (in white) probed by the electronic beam. (B) Molar percentage of Nd, Si, B, Ca and Al along the probed line. Nd, Si, B, Ca and Al molar percentages of the residual glass (far from crystals) are indicated in dashed lines (determined by EMPA).

G0.40Nd8-SC (Fig. 7). The phase separation is even more pronounced at 10 mol% Nd_2O_3 with size of domains of 80 nm for G0.26Nd10-SC glass and of around 330 nm including smaller contrasted droplets of about 20–50 nm for G0.40Nd10-SC (Fig. 7D). Fig. 8 shows the evolution of elements proportion through the dark and bright domains determined by STEM. The dark contrasted phase appears notably enriched in neodymium and slightly depleted in silicon compared to the bright contrasted one. The qualitative estimation of the Nd/Si atomic ratios of the dark and bright contrasted phases was compared to the nominal glass one (0.39), showing a Nd/Si ratio lower than the nominal values for the bright contrasted phase (0.07), and higher for the dark contrasted one (0.51). Furthermore, EFTEM analyses were used to map the distribution of boron and its repartition between the dark and bright zones observed on the G0.40Nd10-SC sample (Fig. 9). According to the boron-filtered image, it was noticed that boron concentration was higher in the dark domains and dark droplets compared to the bright phase. The phase separation observed for the G0.40Nd10-SC sample thus appeared to be constituted of bright neodymium-depleted areas and dark neodymium-boron-rich ones with different domain sizes.

4. Discussion

The behaviors of the quenched and slowly cooled G0.26Nd x and G0.40Nd x glasses based on XRD, SEM and TEM analyses in terms of crystallization and phase separation are summarized in Table 4.

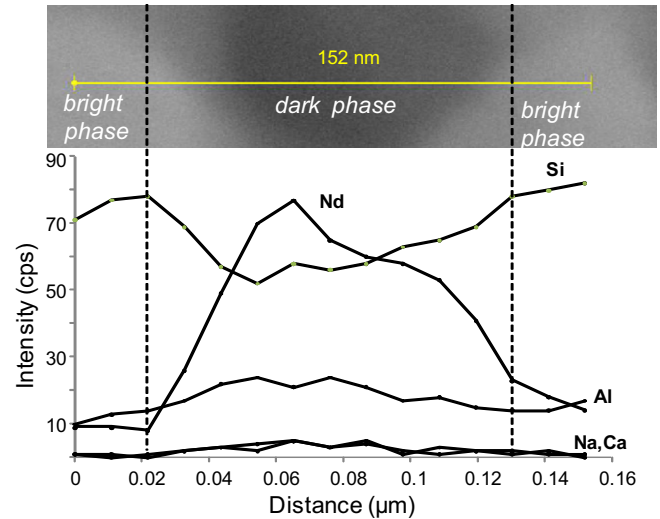


Fig. 8. Element profile measurement performed by STEM EDX analysis of the bright and dark phases along the line for the phase separated zone of the G0.26Nd10-SC sample.

4.1. Homogeneity domains of peraluminous glasses

All the quenched samples for both G0.26Nd x and G0.40Nd x series appear homogeneous for $2.3 \leq x \leq 8$ mol% Nd_2O_3 . Samples also appear completely homogeneous after a slow cooling at 1°C min^{-1} for $4 \leq x \leq 6$ mol% Nd_2O_3 for G0.26Nd x -SC series and for $2.3 \leq x \leq 6$ mol% Nd_2O_3 for G0.40Nd x -SC series. In the side of lower concentrations, some mullite crystallization is observed, meaning that for both series, a minimum Nd_2O_3 amount, inferior to 2.3 mol%, is required to get homogeneous glasses. In the side of upper contents, crystallization and phase separation are observed, indicating a limit of homogeneous incorporation of Nd_2O_3 between 8 and 10 mol% after quenching and between 6 and 8 mol% after a slow cooling at 1°C min^{-1} , whatever the $[\text{B}_2\text{O}_3]/[\text{SiO}_2]$ ratio. Further glasses, with x values ranging between 0 and 2.3 mol% and between 6 and 10 mol% Nd_2O_3 could be tested to precise the limits of these homogeneity domains.

These results highlight the great ability of peraluminous glasses to incorporate homogeneously significant neodymium amounts, without crystallization or phase separation, even for a cooling rate of 1°C min^{-1} . They are in good agreement with DSC analyses and especially with the lack of crystallization peak for $[\text{Nd}_2\text{O}_3] < 8$ mol%, predicting a good thermal stability of these glasses versus temperature.

It is noticeable that the homogeneity domains determined in this work appear much wider than those previously determined by Li et al. for quenched peraluminous glasses (0–4.25 mol% Nd_2O_3), characterized by a R_p and a $[\text{B}_2\text{O}_3]/[\text{SiO}_2]$ ratio close from those of our study [14]. These differences could be attributed to specific glass synthesized routes, as it is well-known that the incorporation limit of an element in a glass strongly depends on its elaboration conditions. Indeed, whereas a

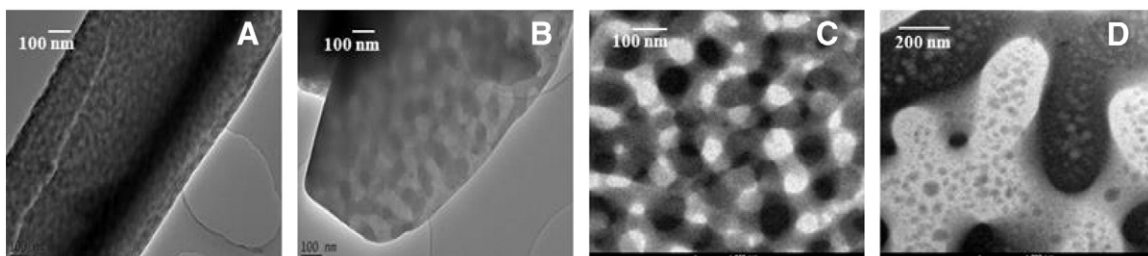


Fig. 7. TEM images of the phase separation observed for (A) G0.26Nd8-SC, (B) G0.40Nd8-SC, (C) G0.26Nd10-SC and (D) G0.40Nd10-SC glasses. Domain sizes are about 20 nm for the G0.26Nd8-SC glass, 80 nm for the G0.40Nd8-SC and G0.26Nd10-SC glasses, and 330 nm including in this case, smaller contrasted droplets of about 20–50 nm.

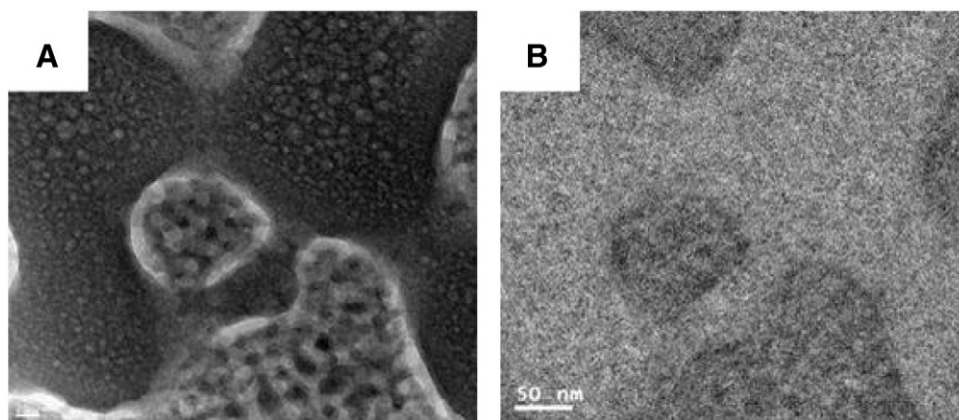


Fig. 9. EFTEM analysis of the G0.40Nd10-SC sample. (A) Zero-loss energy image showing the bright and dark phase separated domains as reference and (B) the corresponding B-filtered image, where bright zones indicate the boron-enriched domains.

two-step elaboration with an intermediary grinding and a total elaboration dwell of 5 h were used in the present study, Li introduced the rare earth oxide into the pre-elaborated glass frit with a step time of 1 h only.

Moreover, it is particularly interesting to note that these homogeneity domains also appear larger than those of peralkaline glasses, for which apatite crystallization was detected above 3.56 mol% Nd_2O_3 after a slow cooling treatment (6°C min^{-1}) [20].

4.2. Phase separation and crystallization tendencies of peraluminous glasses

Interestingly, the two heterogeneity regions from either side of the homogeneity domains appear completely different, concerning the observed crystalline phases and phase separation phenomena.

At low neodymium content, for the G0.26Nd0-SC and G0.26Nd2.3-SC glasses, a mullite-type compound enriched with boron crystallizes. The mullite is well known as solid solution corresponding to the $\text{Al}_2^{VI}\text{O}_2[\text{Al}_2 + 2x\text{Si}_2 - 2x]\text{O}_{10-x}$ general formula, and $0.2 < x < 0.9$ identified in the Al_2O_3 – SiO_2 system and derived from the well-known sillimanite Al_2SiO_5 phase [29,30]. In our case, boron ions are also incorporated in the structure. Fischer et al. suggested that boron ions enter the structure by occupying tetrahedral sites or interstitials and that the charge similarity between Al^{3+} and B^{3+} would favor tetrahedral substitution of Al^{3+} [31]. As no mullite was detected in G0.40Nd2.3-SC, contrary to the G0.26Nd2.3-SC glass, it seems that increasing the $[\text{B}_2\text{O}_3]/[\text{SiO}_2]$ ratio prevents mullite crystallization. Moreover, we observe for both series, that when $[\text{Nd}_2\text{O}_3]$ increases, mullite crystallization disappears at SEM scale. These observations seem to support a synergistic role of boron, aluminum and neodymium in the stabilization of the vitreous network. Actually, we supposed that this could be related to the ability of boron and aluminum to adopt different environments (BO_3 – BO_4 and

AlO_4 – AlO_5 – AlO_6) leading to an extended range of network organization. Moreover, in these peraluminous systems, defined by a lack of charge compensator ions Na^+ or Ca^{2+} regardless to aluminum ions a reasonable hypothesis could consist in estimating Nd^{3+} ions as potential charge compensators of aluminum and/or boron ions. Further structural investigations, in particular using high resolution solid state NMR are actually underway to clarify the charge compensation mode and the role of the different elements.

At high neodymium content (≥ 8 mol% Nd_2O_3), phase separation and/or heterogeneous crystallization can occur after fast quenching or slow cooling (1°C min^{-1}) of the sample.

Phase separation occurred for quenched glasses containing 10 mol% Nd_2O_3 and as early as 8 mol% Nd_2O_3 for slowly cooled glasses. This liquid–liquid phase separation between a bright phase – depleted in neodymium – and a dark contrasted one – enriched in neodymium and boron, and slightly depleted in silicon – looks like a spinodal decomposition rather than a nucleation-growth one.

It can also be noticed that the G0.40Nd10-SC glass presents a phase separation with larger size domains compared to the G0.26Nd10-SC one, suggesting that an increase of the $[\text{B}_2\text{O}_3]/[\text{SiO}_2]$ ratio also seems to promote the phase separation in the network after slow cooling of the glass. This result could probably be linked to the higher concentration of boron in dark domains, highlighting here again the structural key role of boron directly implicated in glass separation in numerous systems [10].

In addition to phase separation, crystallization of a (Nd, Si, B, Ca, O)-phase occurred at high neodymium content, only at the glass–crucible interface for G0.26/0.40Nd8-SC glass and also in the bulk of samples at 10 mol% Nd_2O_3 . The ability of these glasses to crystallize was also revealed by DSC, with the presence of an exothermic crystallization peak for the glasses containing more than 8 mol% Nd_2O_3 . As neodymium content increased, this exothermic peak was shifted

Table 4
Behavior of the quenched and slowly cooled G0.26Nd x and G0.40Nd x glasses based on XRD, SEM and TEM analyses in terms of crystallization and phase separation. Crystalline structure and phase separation are respectively identified by M = mullite crystallization, N = neodymium borosilicate crystallization and P = phase separation. Crystallization/phase separation intensities are indicated in brackets. – corresponds to crystals detected by SEM but not by XRD.

	G0.26Nd0	G0.26Nd2.3	G0.26Nd4	G0.26Nd6	G0.26Nd8	G0.26Nd10
Quenched	M (–)	–	–	–	–	P (20 nm)
Slowly cooled (1°C min^{-1})	M (+++)	M (–)	–	–	N (+) P (20 nm)	N (+++) P (80 nm)
	G0.40Nd0	G0.40Nd2.3	G0.40Nd4	G0.40Nd6	G0.40Nd8	G0.40Nd10
Quenched	–	–	–	–	–	P (20 nm)
Slowly cooled (1°C min^{-1})	M (+++)	–	–	–	N (+) P (80 nm)	N (+++) P (330 nm + 20–50 nm)

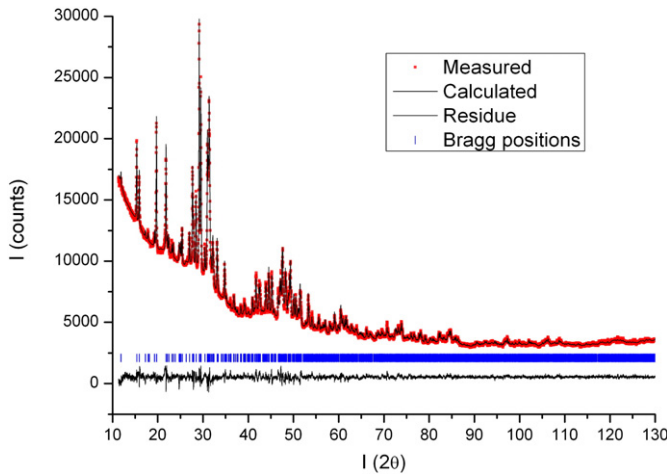


Fig. 10. Rietveld refinement of the X-ray pattern containing the crystalline phase detected in G0.26Nd10-SC sample.

towards lower values, leading to a decrease of $(T_c - T_g)$ and so, to an increasing crystallization tendency of the Nd10 samples, according to the Hruby parameter (value of 0.45 for G0.26Nd10) [32]. Although the B/Si ratios of the crystals found in the G0.26Nd8-SC, G0.26Nd10-SC and G0.40Nd10-SC samples are globally close from each other (0.46–0.58, Table 3), slight variations have to be considered. In the G0.26Nd8-SC and G0.26Nd10-SC samples, the B/Si ratios of the crystals (0.46–0.52) and of the nominal glasses $[B_2O_3]/[SiO_2] = 0.26$ eq. to B/Si = 0.52) are very close, meaning that boron and silicon are incorporated in the crystalline phase in the same proportions as in the glass. For the G0.40Nd10-SC sample, the slight boron enrichment of the crystals composition (B/Si = 0.58) can be related to the higher $[B_2O_3]/[SiO_2]$ of the glass composition $[B_2O_3]/[SiO_2] = 0.40$ eq. to B/Si = 0.81).

Furthermore, even if no crystalline phase containing these elements could be identified in structural databases, the indexations of the diagram match with an Europium-rich phase, of orthorhombic structure and of general formula $Eu_3Si_2BO_{10}$ (ICCD-card: 86-1296) (space group: Pbc_a). As the EMPA results show that the mean crystalline composition presents a Nd/Si/B stoichiometry close to 3/2/1, an hypothetical phase 'Nd₃ + xSi₂ + yB₁ + zO₁₀ + δ' could be proposed, with a small calcium/neodymium substitution and with a B/Si ratio slightly varying with the one of the glass compositions.

A Rietveld analysis was carried out on crystals detected in G0.26Nd10-SC glass to describe more precisely this crystalline structure (Fig. 10 and Table 5). Cell parameters were extracted from the refinement and compared with those of other RE₃Si₂BO₁₀ or

RE₃Si_{0.92}Ge_{1.08}BO₁₀ orthorhombic structures referenced in the literature (with RE = Eu, Sm, Nd, Ce, La) [33–37] (Table 6). These parameters appear very close from those of the literature, with values situated between those of Ce₃Si₂BO₁₀ and Sm₃Si₂BO₁₀, in agreement with the evolution of ionic radii of the respective rare earth incorporated in the different phases [38]. The good agreement of the pattern refinement with an isostructural model from the RE₃Si₂BO₁₀ family by the Rietveld technique strongly supports the formation of a Nd₃ + xSi₂ + yB₁ + zO₁₀ + δ phase slightly substituted by Ca (and perhaps Al at very low level).

Thus, crystallization of a phase close to Nd₃Si₂BO₁₀, and phase separation between a (Nd,B)-enriched dark phase, slightly depleted in silicon, and a bright phase depleted in neodymium were both observed at high neodymium content. The question is now to understand if the crystallization process is induced by the phase separation, with crystals growing probably from the dark areas or if the two phenomena occur independently, simultaneously or at different time scales.

As for molybdenum-rich borosilicate glasses [39,40], a phase separation could occur at $T_p < T_{elab}$ (T_p = phase separation temperature, T_{elab} = elaboration temperature), followed by the crystallization of phases located in separated areas at $T_c < T_p$ (T_c = crystallization temperature). In such glasses, the liquid–liquid phase separation occurs following a nucleation–growth mechanism and leads to dark droplets, mostly enriched in Mo and alkaline/alkaline-earth elements, and to a residual vitreous phase depleted in Mo. If the quenching rate of the glass is fast enough, the glass can be homogeneous. At smaller cooling rate, the glass will be phase separated and probably crystallized, depending on the cooling rate.

In our system, if the crystallization process would have arisen from the dark areas, a crystalline phase enriched in neodymium and boron and slightly depleted in silicon compared to the residual matrix should have been detected. According to EMPA results, the crystalline phases of G0.26Nd8/10-SC and G0.40Nd10-SC glasses are effectively strongly enriched in neodymium and slightly depleted in silicon compared to the respective nominal glass, but are also slightly depleted in boron. Another hypothesis could be that separated phases could act as nucleation points for these crystals.

Therefore, in our systems, phase separation and crystallization processes can both take place in glasses containing above 8 mol% Nd₂O₃, with a respective proportion depending on the cooling rate. Nevertheless, further investigations are required to better understand the crystallization/phase separation mechanisms of these glasses.

Finally, from these results, it seems that the microstructure of our glasses shows differences with those of other peraluminous glasses previously studied [10–14]. First of all, even if STEM analyses revealed a phase separation above 8 mol% Nd₂O₃ in our glasses, it seems that it is notably controlled by a specific neodymium distribution, particularly concentrated in the dark domains. Profile measurements also revealed that both dark and bright contrasted phases contained silicon, meaning that the phase separation does not imply separated borate and silicate portions, as proposed by Li [10–14]. Secondly of all, no apatite structure was detected in our samples above the rare-earth solubility limit, contrary to other peraluminous [10–14] or peralkaline glasses [4,15–17, 20], for which NaRE_{9.33}(SiO₄)₆O₂ or Ca₂Nd₈(SiO₄)₆O₂ apatite structures containing high proportions of alkaline or alkaline-earth elements were detected.

5. Conclusion

In this study, homogeneity of peraluminous glasses was investigated at various scales, versus the neodymium content and the $[B_2O_3]/[SiO_2]$ ratio, at different cooling rates. Quenched peraluminous samples are homogeneous for $2.3 \leq x \leq 8$ mol% Nd₂O₃ as well as slowly cooled glasses ($1^\circ \text{C min}^{-1}$) for $2.3-4 \leq x \leq 6$ mol% Nd₂O₃, evidencing the large homogeneity domain of such glass compositions, even more extended for G0.40Nd_x series. Even if this latter series seems to be the

Table 5

Atomic positions extracted from Rietveld refinement of the X-ray pattern containing the crystalline phase detected in G0.26Nd10-SC sample.

Atom	x	y	z
Nd1	0.1670(8)	0.1613(18)	0.3636(4)
Nd2	0.3931(7)	0.3184(13)	0.2404(4)
Nd3	0.1397(8)	0.07382(17)	0.0081(4)
Si1	0.147(4)	0.4279(7)	0.100(2)
Si2	0.146(4)	0.2121(10)	0.063(2)
B	0.384(10)	0.008(3)	0.372(6)
O1	0.145(5)	0.2966(12)	0.076(3)
O2	0.133(6)	0.3876(10)	0.240(3)
O3	0.196(4)	0.2795(11)	0.408(3)
O4	0.144(6)	0.5124(14)	0.160(3)
O5	0.413(4)	−0.0580(13)	0.336(3)
O6	0.326(5)	0.4083(12)	0.032(3)
O7	0.329(5)	0.2079(11)	0.166(3)
O8	0.028(5)	0.1705(15)	0.111(3)
O9	0.299(4)	0.0619(13)	0.269(3)
O10	0.591(5)	0.4272(13)	0.520(4)

Table 6
Cell parameters (a, b, c and cell volumes) of orthorhombic RE₃Si₂BO₁₀ or RE₃Si_{0.92}Ge_{1.08}BO₁₀ single crystals referenced in the literature (with RE = La, Nd, Ce, Sm, Eu) (space group: Pbca) and compared to one of the crystals present in the G0.26Nd10-SC sample, extracted from the whole pattern decomposition.

	a (Å)	b (Å)	c (Å)	V (Å ³)	Ionic radii RE ³⁺		Ref
					CN = 8	CN = 9	
Eu ₃ Si ₂ BO ₁₀	9.673(1)	6.992(3)	22.867(3)	1547(1)	1.12	1.07	[33]
Sm ₃ Si ₂ BO ₁₀	9.683(2)	7.018(2)	22.904(5)	1556(1)	1.13	1.08	[34]
Nd-crystals in the G0.26Nd10-SC	9.78(6)	7.09(6)	23.08(1)	1600(24)	1.16	1.11	This study
Nd ₃ Si _{0.92} Ge _{1.08} BO ₁₀	9.845(1)	7.146(1)	23.382(2)	1645(1)			[35]
Ce ₃ Si ₂ BO ₁₀	9.900(7)	7.203(6)	23.292(2)	1661(3)	1.20	1.14	[36]
La ₃ Si ₂ BO ₁₀	9.977(2)	7.269(2)	23.410(2)	1698(1)	1.22	1.16	[37]

more interesting in terms of homogeneity, glasses have also to be studied in terms of long term behavior, since the boron content is significantly higher than in G0.26Nd_x series.

Compared to peralkaline compositions, peraluminous glasses present a higher rare earth solubility rate after slow cooling treatment, meaning that a large increase of the alumina content in regard of modifier oxides ones, enhances neodymium structural distribution, contributing to a better glass stability. As minimal rare earth content is also necessary to prevent mullite crystallization, it seems that the rare earth also enhances the alumina stability in the network and probably its structural distribution. At higher content in Nd₂O₃ (>8 mol%), phase separation phenomena between a (Nd,B)-enriched phase and a phase depleted in neodymium, and crystallization have been observed. In particular, a neodymium enriched phase Nd₃+_xSi₂+_yB₁+_zO₁₀+_δ has been identified. For a better understanding of the system, further structural investigations including high resolution solid-state NMR are therefore in progress to determine the aluminum and the rare earth structural roles in the peraluminous network.

Acknowledgments

The financial support by the CEA Marcoule (Commissariat à l'Energie Atomique, France) and by Areva (France) for this work is gratefully acknowledged. This work takes part of a project coordinated by the International Atomic Energy Agency (IAEA) that the authors would also like to acknowledge.

References

- [1] J.-M. Gras, R.D. Quang, H. Masson, T. Lieven, C. Ferry, C. Poinssot, M. Debes, J.-M. Delbecq, J. Nucl. Mater. 362 (2–3) (2007) 383.
- [2] P. Frugier, C. Martin, I. Ribet, T. Advocat, S. Gin, J. Nucl. Mater. 346 (2005) 194.
- [3] C. Lopez, X. Deschanel, J.M. Bart, J.M. Boubals, C. Den Auwer, E. Simoni, J. Nucl. Mater. 312 (2003) 76.
- [4] A. Kidari, J.-L. Dussossoy, E. Brackx, D. Caurant, M. Magnin, I. Bardez-Giboire, J. Am. Ceram. Soc. (2012) 1.
- [5] E.M. Erbe, D.E. Day, J. Am. Ceram. Soc. 73 (1990) 2708.
- [6] J.E. Shelby, J.T. Kohli, J. Am. Ceram. Soc. 73 (1) (1990) 39.
- [7] R.K. Brow, D.R. Tallant, G.L. Turner, J. Am. Ceram. Soc. 80 (11) (1990) 39.
- [8] M.S. Spess, J.E. Shelby, Phys. Chem. Glasses 33 (3) (1982) 87.
- [9] H.L. Rutz, D.E. Day, C.F. Spencer Jr., J. Am. Ceram. Soc. 73 (6) (1985) 1788.
- [10] L. Li, D.M. Strachan, H. Li, L.L. Davis, M. Qian, J. Non-Cryst. Solids 272 (2000) 1.
- [11] L. Li, H. Li, M. Qian, D.M. Strachan, J. Non-Cryst. Solids 283 (2001) 237.
- [12] L. Li, H. Li, M. Qian, D.M. Strachan, J. Non-Cryst. Solids 281 (2001) 189.
- [13] M. Qian, L. Li, H. Li, D.M. Strachan, J. Non-Cryst. Solids 333 (2004) 1.
- [14] H. Li, L. Li, D.M. Strachan, M. Qian, J. Non-Cryst. Solids 349 (2004) 127.
- [15] D. Caurant, A. Quintas, O. Majerus, T. Charpentier, I. Bardez, Adv. Mater. Res. 39–40 (2008) 19–24.
- [16] A. Quintas, O. Majerus, D. Caurant, J. Am. Ceram. Soc. 90 (3) (2007) 712.
- [17] A. Quintas, D. Caurant, O. Majerus, Phys. Chem. Glasses: Eur. J. Glass Sci. Technol. B 49 (4) (2008).
- [18] A. Quintas, D. Caurant, O. Majerus, T. Charpentier, J.-L. Dussossoy, Mater. Res. Bull. 44 (2009) 1895.
- [19] I. Bardez, D. Caurant, P. Loiseau, N. Baffier, Phys. Chem. Glasses 46 (4) (2005) 320.
- [20] I. Bardez, D. Caurant, J.-L. Dussossoy, P. Loiseau, C. Gervais, F. Ribot, D. Neuville, N. Baffier, C. Fillet, Nucl. Sci. Eng. 153 (2006) 272.
- [21] I. Bardez, D. Caurant, F. Ribot, P. Loiseau, J.-L. Dussossoy, F. Villain, N. Baffier, C. Fillet, Mater. Res. Soc. Symp. Proc. 807 (2003) 157.
- [22] D. Caurant, O. Majerus, P. Loiseau, I. Bardez, N. Baffier, J.-L. Dussossoy, J. Nucl. Mater. 354 (2006) 143.
- [23] L. Li, M. Qian, H. Li, D.M. Strachan, Ceram. Trans. 107 (2000) 131.
- [24] J. Rodriguez-Carvajal, Phys. B Condens. Matter 192 (1993) 55–69.
- [25] A. Le Bail, Mater. Res. Bull. 23 (1988) 447.
- [26] H.M. Rietveld, J. Appl. Cryst. 2 (1969) 65.
- [27] O. Gedeon, K. Jurek, V. Hulinsky, J. Non-Cryst. Solids 246 (1999) 1–8.
- [28] J.E. Shelby, Rare elements in glasses, in: J.E. Shelby (Ed.), Key Engineering Materials, 94–95, Trans Tech Publications, 1994, p. 43.
- [29] I. Jaymes, A. Douy, D. Massiot, J.-P. Coutures, J. Non-Cryst. Solids 204 (1996) 12.5.
- [30] H. Schneider, J. Schreuer, B. Hildmann, J. Eur. Ceram. Soc. 28 (2008) 329.
- [31] R.X. Fischer, H. Schneider, in: H. Schneider, S. Komarneni (Eds.), Mullite, Wiley-VCH Verlag GmbH and Co., KGaA, Weinheim, 2005, p. 90.
- [32] A. Hrubý, Czech J. Phys. B22 (1972) 1187–1193.
- [33] L. Chi, et al., Acta Crystallogr. C52 (1996) 2385–2387.
- [34] L. Chi, et al., J. Alloys Compd. 242 (1996) 1–5.
- [35] K. Serhan, et al., J. Solid State Chem. 110 (1994) 384–388.
- [36] H. Müller-Bunz, et al., Z. Anorg. Allg. Chem. 627 (2001) 1436–1438.
- [37] E.V. Shvanskii, et al., J. Solid State Chem. 154 (2000) 312–316.
- [38] R.D. Shannon, Acta Crystallogr. A32 (1976) 751.
- [39] O.V. Mazurin, G.P. Roskova, E.A. Porai-Koshits, in: O.V. Mazurin, E.A. Porai-Koshits (Eds.), Phase Separation in Glass, Elsevier Science Publishers B.V., Amsterdam, 1984.
- [40] C. Cousi, F. Bart, J. Phalippou, Glass Technol. 45 (2004) 65.